FairTransCP: Live Demo Presentation Guide

Step-by-Step Oral Script & Walkthrough for Faculty Review (AISTATS 2027 Submission)

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Live Demo: fairtranscp-demo.onrender.com	**GitHub:** github.com/aarushdubey/fair-conformal-finance

Step 0: The 1-Minute Pre-Meeting Checklist

- **Wake Up Render:** Open fairtranscp-demo.onrender.com 2 minutes prior so the server is awake and active.
- **Local Backup:** Keep http://localhost:8501 running locally on your laptop as zero-latency insurance.
- **Maximized Screen:** Keep your browser maximized so all cards and side-by-side comparison boxes fit comfortably.

Step 1: The 30-Second Opening Statement

(Before clicking anything on screen, look at your professor and set the real-world context:)

> *"Professor, in financial lending—such as personal loans or credit lines—banks rely on machine learning models. However, standard models output only a binary point prediction: **Approve** or **Reject**.*
>
> *If an applicant has a 51% default risk, standard ML rejects them identically to someone with a 99% risk, concealing the model's true uncertainty. Under the Equal Credit Opportunity Act (ECOA) and the EU AI Act, uncalibrated uncertainty exposes banks to severe regulatory liability.*
>
> *In this project, we apply **Conformal Prediction** to produce prediction sets with a mathematically guaranteed 90% confidence level. However, when we enforced fairness across demographic groups, we uncovered an unexpected failure mode: **forcing equal coverage actively harms minority borrowers**. We formalized this as the **Coverage-Equity Paradox**, and we designed **FairTransCP** to solve it. Let me show you on our live dashboard."*

Step 2: Tab 1 — ■ The Paradox & Core Concept

Action: Navigate to Tab 1. Point to the three colored cards at the top.

What to Say:

> *"Here are the three paradigms:*
> *1. **Standard CP (Blue):** Computes a single population threshold. It achieves 90% coverage overall, but historically underrepresented groups (e.g. young borrowers) suffer severe undercoverage (e.g. 81%).*
> *2. **Mondrian Group CP (Orange):** The standard academic fix. It enforces equal 90% coverage by computing separate thresholds for each demographic group.*
> *3. **The Catch (The Paradox):** Because minority groups have higher variance scores, equalizing coverage forces the algorithm to push that group's threshold significantly higher.*
>
> *If you look at the curve below, we proved in **Proposition 1** that the ambiguous zone width is **W(q) = max(0, 2q - 1)**. As the threshold rises, the ambiguous probability band expands linearly! The model gives up and outputs **{Approve, Reject}**. In retail banking, those borrowers get shunted to manual underwriting queues and administrative friction."*

Step 3: Tab 2 — ■ Live Applicant Simulator

Action: Click Tab 2. Demonstrate two specific applicants to highlight the exact contrast.

Applicant	Predicted Prob	Standard CP	Mondrian CP	FairTransCP (Ours)
Applicant #2 Young (<25 yrs)	55.9% (Good)	{Bad, Good}	{Bad, Good} *(Ambiguous)*	{Bad, Good} *(Valid boundary case)*
Applicant #26 Standard (>=25 yrs)	28.1% (Good)	{Bad, Good} *(Indecisive)*	{Bad, Good} *(Ambiguous!)*	**{Bad}** **Confident Decision!**

What to Say:

> *"On **Applicant #2** (a young borrower near the 55% boundary), all models output {Bad, Good} reflecting true statistical uncertainty.*
>
> *Now look at **Applicant #26**: Both Standard CP and Mondrian CP output {Bad, Good} (indecisive). Under the Fair Credit Reporting Act (FCRA), an ambiguous set cannot support an automated Adverse Action notice. **FairTransCP** regularizes the group threshold, cutting through the excess ambiguity and outputting a decisive **{Bad}** without compromising coverage safety."*

Step 4: Tab 3 — ■ Empirical Benchmark Results

Action: Click Tab 3. Select **Dataset:** Adult Income (UCI) and **Model:** XGBoost.

What to Say:

> *"We benchmarked FairTransCP across 3 real-world datasets—German Credit, Taiwan Credit, and Adult Income—using strictly classical tree ensembles (Random Forest, XGBoost, LightGBM).*
>
> *Look at the Adult Income + XGBoost benchmark:*
> *• On the left chart, Mondrian CP equalizes coverage across demographics.*
> *• But on the right chart, **Set-Size Disparity surges to 1.637 (+57.4% explosion!)**. Female applicants receive over 1.6 times more ambiguous sets than male applicants.*
> *• **FairTransCP** dampens this disparity down to **1.040** while maintaining 95.3% worst-group coverage.*
>
> *(If asked about Generic Fair CP): Generic Fair CP achieves disparity of 1.000 trivially by outputting set size 2.000 for every single person. Outputting all classes is useless in commercial banking."*

Step 5: Tab 4 — ■■ The λ Pareto Dial & Theoretical Bounds

Action: Click Tab 4. Move the **Interpolation Weight (λ)** slider on the sidebar from 0.0 to 1.0.

What to Say:

"FairTransCP interpolates group and pooled quantiles via:
***q■^(g)(λ) = (1 - λ) q■^(g) + λ q■_pooled**, where λ ∈ [0, 1].*

As I drag λ across the slider:
• At λ = 0: Pure Mondrian (zero coverage gap, maximum disparity explosion).
• At λ = 1: Pure Standard CP (lowest set sizes, but demographic coverage gaps).
• Between 0.3 and 0.5: The balanced operating regime on the Pareto frontier.

*Importantly, in **Theorem 1**, we prove a distribution-free coverage lower bound: the coverage loss is strictly bounded by the probability mass between the two quantiles, scaled down by λ."*

Step 6: Professor Defense Cheatsheet & Tough Questions

Keep Tab 5 open on screen as reference while answering questions:

Question	Exact 1-2 Sentence Defense Answer
Why not just use Mondrian CP?	“Mondrian equalizes coverage percentages, but it ignores prediction set sizes. It produces 1.6x larger, ambiguous sets for protected groups, replacing algorithmic error with administrative bias.”
Why no Deep Learning or Transformers?	“Under FCRA adverse action rules and the EU AI Act, retail credit scoring models must provide clear feature-attribution explanations. Tree ensembles (XGBoost, LightGBM) are the actual production standard in banking.”
Why does FairTransCP match Standard CP on Taiwan Credit?	“Because Taiwan Credit is already balanced across genders; our dual-objective loss automatically sets $\lambda^* \approx 1.0$, gracefully recovering standard CP rather than introducing unnecessary distortions.”
What prevents arbitrary coverage collapse?	“Theorem 1 establishes a formal distribution-free lower bound proving that coverage loss is strictly bounded by the probability mass between quantiles, scaled down by λ.”

Conclusion & Wrap-Up

Wrap up your presentation confidently with:
“Professor, all experiments, automated unit tests, and the demo application are open-source on GitHub. I would value your guidance on positioning our Pareto trade-off and Theorem 1 bound for our target AISTATS 2027 submission.”